

Gürkan Doğukan Karakurt

CIVIL ENGINEER · SOFTWARE DEVELOPER · GRADUATE RESEARCHER

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Research Assistant @ METU Civil Engineering — Construction Management. Civil engineer with a double major in Corporate Finance, combining structural engineering with full-stack software development (Python/Django, React, React Native) and applied graphic design. Research interests: Construction Management, BIM, Machine Learning.

EDUCATION

2025 – Present	MSc, Construction Management Middle East Technical University (METU), Ankara Graduate research in Construction Management with a focus on BIM-driven planning and machine-learning approaches for construction analytics. Transferred from the Geotechnical Engineering programme in Spring 2025.
2017 – 2023	BSc, Civil Engineering Middle East Technical University (METU) — GPA 3.35 / 4.00 Coursework in structural analysis, geotechnics, hydraulics, construction management, and Building Information Modeling.
2019 – 2023	BSc Double Major, Corporate Finance Middle East Technical University (METU) — GPA 3.39 / 4.00 Financial analysis, valuation, and quantitative methods alongside the civil engineering programme.

EXPERIENCE

Oct 2025 – Present	Research Assistant — Construction Management Middle East Technical University (METU), Department of Civil Engineering Graduate research and teaching support within the Construction Management division. Focus areas include BIM-driven planning, construction analytics, and machine learning.
Dec 2024 – Oct 2025	Software Engineer Mega Mühendislik, Türkiye Developed software for engineering workflows, contributing across the stack to ship reliable internal tooling before moving full-time into graduate research at METU.
2023 – 2024	Head of Quantitative Strategies Vortex · Remote (Georgia) Designed and developed trading bots for financial markets. Built scalping and swing strategies, refined algorithm performance, and tailored bots to investor-specific mandates so the startup could offer customised allocation services.
2021 – 2024	Web Development Project Lead Self-initiated project Led a self-started web development project focused on financial markets, delivering a web-based trading interface end to end. Coordinated tasks across the project, owned product decisions, and extended the startup's capability around investment management.
Summer 2022	Civil Engineering Intern ES Group — Erdemir Port Project, Zonguldak Worked across site operations, planning, and progress payments on a port construction project. Contributed to steel-pile design and took responsibility in cost control and project scheduling, bridging engineering principles with field execution.
2020 – 2024	Freelance Graphic Designer · 2024 Local Elections Lead Designer Independent Delivered posters, brochures, social media content, and campaign materials across freelance projects. In the 2024 Turkish local elections, served as lead designer for a mayoral campaign and owned the full digital content production pipeline.

FEATURED PROJECT

Diyet Cebimde — AI-powered diet & nutrition platform — full-stack, solo-built
2025 — Ongoing · Sole Full-Stack Developer / Founder · Mobile + Web Admin + API
A full-stack mobile platform that generates AI-driven weekly diet plans from each user's personal profile and health goals. Features photo-based meal recognition, manual nutrition tracking, weight and body-measurement logging, and a memory-equipped chatbot. I designed and built the entire system end to end — backend API, admin web panel, and the React Native mobile client.
Engineering highlights: hybrid AI + linear programming pipeline (Gemini for meal selection, PuLP for portion optimisation), snapshot architecture so historical plans never break when the catalog changes, four-tier model fallback strategy for rate-limit resilience, 34-field central food model with row-level CSV import savepoints, six parallel-loaded analytics dashboards.
Stack: Django 5.2 · DRF · PostgreSQL · SimpleJWT · Google OAuth · Gemini API · PuLP (LP) · React 18 + TS · Vite · Tailwind · React Query · Zustand · Expo SDK 53 · React Native · NativeWind · i18next · Railway · EAS Build

SKILLS

Software Development: Python & Django (advanced), HTML / CSS / JS (advanced), React (intermediate), PostgreSQL (intermediate), MATLAB, Visual Basic (advanced), Financial algorithms (expert)

Graphic Design: Photoshop (advanced), Illustrator (advanced), Premiere Pro (advanced), After Effects (intermediate)

Civil Engineering: AutoCAD (upper-intermediate), Excel + VBA (expert), SAP2000 (beginner-intermediate), EPANET (advanced), MS Project (intermediate)

Languages: Turkish (native), English (C1)

Personal contact details (email, phone, address) are intentionally omitted from this CV for privacy. Please reach out through LinkedIn or ORCID for academic correspondence.